Albin Zeqiri

PhD Candidate in Human–Computer Interaction, Ulm University

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Profile

I am a **final-year PhD candidate in Human–Computer Interaction** at Ulm University. Using my background in machine learning and experimental research methods, **I study how interactive systems can help users reach their intended goals while meeting ecological objectives set by companies and policymakers**. Leveraging a use-case-based research approach, I have studied this across contexts such as mobility, appliances, and urban robotics. My current work focuses on building design, where I use XR and generative AI to support the planning and use of spaces. In collaborations across industry and academia (**Mercedes-Benz Tech Innovation, UCL Interaction Centre, Karlsruhe Institute of Technology**), I was responsible for planning, implementing, and evaluating customized machine learning pipelines. My research and collaborative projects have been published at top-tier venues such as **CHI** and **IMWUT**.

Skills

Experimental Design: Online, laboratory, and field-based experiments (exploratory and hypothesis-driven); large-scale literature surveys (e.g., PRISMA); data mining using multi-API data collection and NLP-based analysis
Analysis: Statistical analysis; machine learning and deep learning (supervised and self-supervised); qualitative analysis using thematic analysis and grounded theory
Generative & Agentic AI: LLM APIs (OpenAI, Anthropic, Llama); LLM agents with tool use; diffusion models (Stable Diffusion, ComfyUI); agentic coding (Claude Code, Cursor, GitHub Copilot)
Programming Languages: Python, C#, JavaScript, R, SQL, C/C++, Java, Bash, HTML, CSS
Frameworks and Tools: PyTorch, TensorFlow, CUDA, OpenCV, Unity, React, Node.js, Flask, Git, PostgreSQL
Design: UI/UX prototyping and wireframing; image and video editing (Adobe Creative Cloud)
Languages: Albanian (native), German (native), English (fluent), French (intermediate)

Work Experience

Ulm University – Chair of Human–Computer Interaction 09/2022 – present
Research Associate

- Studied how interactive systems can meet ecological goals without hurting user experience through experiments, field deployments of artifacts, and interviews, resulting in **three first-author full papers at CHI and IMWUT**.
- Independently initiated and maintained a collaboration with the Botanical Garden Ulm, leading the development of a robot-based remote-gardening system and a mobile visitor app to evaluate design concepts with real users (**CHI '25, CHI '26**).
- Built reproducible deep learning pipelines, ablation studies, and multi-API data mining workflows for **UCL Interaction Centre**, **KIT**, and **Mercedes-Benz Tech Innovation**, leading to three joint publications.
- Co-authored a successful **€1,250,000 DFG Reinhart Koselleck proposal** on everyday XR for better building utilization, which frames my current building-design work.
- Supervised 40+ students from problem formulation to publication; **two student projects were published at CHI** and two students continued to a PhD.

Ulm University – Chairs of Human–Computer Interaction and Visual Computing 12/2019 – 04/2022
Research Assistant and Tutor

- Built VR prototypes in Unity and C# and supported the user studies run with them, giving the group working testbeds for new interaction techniques.
- Benchmarked monocular depth estimation models for accuracy and scalability to inform model selection.
- Taught C#, Unity, and UI/UX prototyping in tutorials.

Education

PhD in Human–Computer Interaction 09/2022 – 12/2027 (expected)
Ulm University
Dissertation: Between User Experience and Planetary Limits: Designing Interactive Systems with Ecological Objectives
Advisor: Prof. Dr. Enrico Rukzio

M.Sc. Computer Science 03/2020 – 08/2022
Ulm University Grade: 1.3 (A-equivalent)
Thesis (1.0, A-equivalent): A Dataset and Temporal Modeling Approach for Automated Thermal Comfort State Recognition, **published in IMWUT '24** (doi:10.1145/3678503)

B.Sc. Computer Science 10/2016 – 12/2019
Ulm University Grade: 2.0 (B-equivalent)
Thesis (1.0, A-equivalent): Depth Levels: Measuring Achievable Levels of Voluntary Vergence Eye Movements for Eye-based Human–Computer Interaction

Publications

Full Papers

CHI '26 **Investigating the Effects of Eco-Friendly Service Options on Rebound Behavior in Ride-Hailing**
A. Zeqiri, M. Rietzler, E. Rukzio · doi:10.1145/3772318.3790711
Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)

CHI '25 **PlantPal: Leveraging Precision Agriculture Robots to Facilitate Remote Engagement in Urban Gardening**
A. Zeqiri, J. Britten, C. Schramm, P. Jansen, M. Rietzler, E. Rukzio · doi:10.1145/3706598.3713180
Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)

CHI '25 **Bumpy Ride? Understanding the Effects of External Forces on Spatial Interactions in Moving Vehicles**
M. Sasalovici, **A. Zeqiri**, R. C. Schramm, O. J. Ariza Nuñez, P. Jansen, J. P. Freiwald, M. Colley, C. Winkler, E. Rukzio · doi:10.1145/3706598.3714077
Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)

IMWUT '24 **'Eco Is Just Marketing': Unraveling Everyday Barriers to the Adoption of Energy-Saving Features in Major Home Appliances**
A. Zeqiri, P. Jansen, J. O. Rixen, M. Rietzler, E. Rukzio · doi:10.1145/3643558
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), Vol. 8, No. 1

IMWUT '24 **AutoTherm: A Dataset and Benchmark for Thermal Comfort Estimation Indoors and in Vehicles**
M. Colley*, S. Hartwig*, **A. Zeqiri**, T. Ropinski, E. Rukzio · doi:10.1145/3678503
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), Vol. 8, No. 3

Late-Breaking Work and Short Papers

CHI '26 **BotaXplore: Enhancing Visitor Engagement and Learning in Botanical Gardens Through Mobile Technology**
A. Zeqiri*, T. Wagner*, J. Grüneberg, E. Rukzio · doi:10.1145/3772363.3799272
Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)

UbiComp '24 **Exploring Contextual Feature Combinations for Prediction of Subjective Thermal Perceptions**
A. Zeqiri, M. Rietzler, E. Rukzio · doi:10.1145/3675094.3678487
Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)

* equal contribution

Teaching

Interactive Visual Design 10/2022 – present

Co-Organizer, Institute of Media Research and Media Development

Run the seminar series on data visualization libraries such as D3.js, and mentor student groups one-on-one as they design and build their own interactive visualization projects.

Visual Design 10/2022 – present

Co-Organizer, Chair of Media Research and Media Development

Built an out-of-the-box web framework that lets students with little web development experience take part, and teach an introductory seminar series on responsive web design with HTML, CSS, and JavaScript libraries.

Research Project: User-Centred Design 10/2022 – present

Co-Organizer, Chair of Human–Computer Interaction

Co-organize and supervise interdisciplinary, year-long research group projects built on user-centered design and design thinking.

Seminar: Research Trends in Media Informatics 10/2022 – 02/2024

Co-Organizer, Chair of Human–Computer Interaction

Taught students one-on-one how to conduct literature surveys with the PRISMA method, and took part in assessment and grading.

Student Supervision

Supervised 40+ theses and research projects. Selected students:

- Julian Britten (Ulm University and Botanical Garden Ulm, 2024; now PhD student at Ulm University) – **CHI '25 full paper** (doi:10.1145/3706598.3713180)
- Johanna Grüneberg (Ulm University and Botanical Garden Ulm, 2024) – **CHI '26 late-breaking work** (doi:10.1145/3772363.37
- Omid Rajabi (Ulm University, 2026; now PhD student) – **full paper under review**
- Benno Hölz (Ulm University, 2026)
- Christine Mayer (Ulm University and Exxcellent Solutions GmbH, 2025)
- Johannes Martin Ertle (Ulm University, 2024)

Scholarly Service & Outreach

Peer Review: 50+ manuscripts for CHI ('23–'26), IMWUT ('23–'25), IUI ('24, '26), DIS ('24), IEEE VR ('23, '24), HRI ('25), AutoUI ('24, '25), CHI PLAY ('23, '25), PACMHCI, C&T ('25), and MuC ('24)

Outstanding Review Recognitions (10): CHI (4x), IMWUT (3x), DIS, MobileHCI, and MuC, 2024–2026

AI Grid Member (now Alumnus) 08/2023

Selected through competitive review for AI Grid, a network of early-career AI researchers in Germany funded by the Federal Ministry of Education and Research, with a focus on interdisciplinary exchange and responsible AI research

AR Experience for *Spaceman*, Theater Ulm 09/2026

Co-developed the passthrough AR experience for the production, in which virtual scenery and characters interact with the actors throughout the multi-hour play

Invited Workshop, PRIO1 Climate Event, Heidelberg 11/2026

Leading the workshop *Roboter im Stadtbeet?* on future scenarios for robot-supported urban gardening, building on my CHI '25 paper

Fellowships & Grants

DFG Reinhart Koselleck Project *VRooms* (€1,250,000), co-author 2025
Co-authored the successful proposal with Prof. Dr. Enrico Rukzio: *Fighting Climate Change by Increasing the Utilization of Buildings through Everyday Extended Reality* (project page)

Landesgraduiertenförderung (LGFG) Doctoral Fellowship 2023
Competitive three-year fellowship funding my PhD research

DAAD Travel Grant 2026
German Academic Exchange Service grant to present at CHI ’26

Graduate Travel Grant (€2,800) 2024
Graduate & Professional Training Center Ulm grant to present two papers at UbiComp ’24